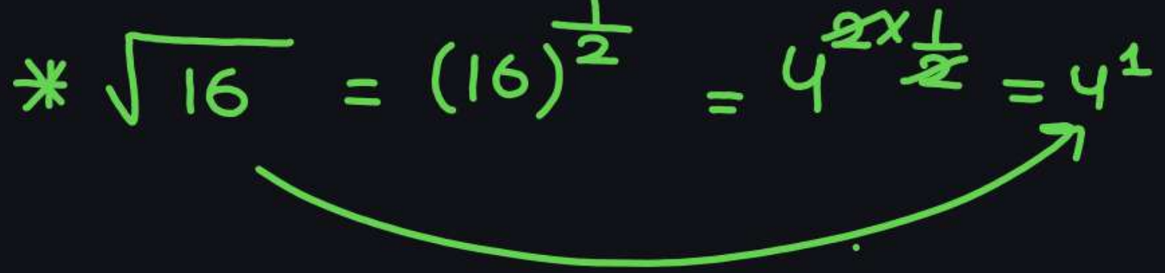
SURDS & INDICES

घातांक और करणी

CLASS NOTES

TYPE – 01

BASIC QUESTIONS


* √16 = (16)^1/2 = 4^2x1/2 = 4^1

1. If $x = \sqrt{64} + \sqrt{121} - \sqrt{169}$, then find the value of x^2.

यदि $x = \sqrt{64} + \sqrt{121} - \sqrt{169}$ है, तो x^2 का मान ज्ञात करें।

SSC CGL TIER- II 07/03/2023

(a) 16
(b) 25
(c) 36
(d) 49

$x = 8 + 11 - 13$
$= 6$

$\therefore x^2 = 6^2 = 36$

2. Find the value of

$$\sqrt{625}+\sqrt{\frac{625}{100}}+\sqrt{\frac{625}{10000}}+\sqrt{\frac{625}{1000000}}$$

$$\sqrt{625}+\sqrt{\frac{625}{100}}+\sqrt{\frac{625}{10000}}+\sqrt{\frac{625}{1000000}}$$ का मान ज्ञात कीजिए।

SSC CHSL MAINS 10/01/2024 (Shift-01)

(a) 27.025 (b) 27.775

(c) 27.075 (d) 27.75

$25+\frac{25}{10}+\frac{25}{100}+\frac{25}{1000}$

$= 25+2.5+0.25+0.025$

$= 27.775$

2 | 0.025
2 | 0.25
2 | 2.5
27.775

$25+\frac{25}{10}+\frac{25}{100}+\frac{25}{1000}$

$= 25+2.5+0.25+0.025$

$= 27.775$

2. Find the value of

$\sqrt{625}+\sqrt{\frac{625}{100}}+\sqrt{\frac{625}{10000}}+\sqrt{\frac{625}{1000000}}$

$\sqrt{625}+\sqrt{\frac{625}{100}}+\sqrt{\frac{625}{10000}}+\sqrt{\frac{625}{1000000}}$ का मान ज्ञात कीजिए।

SSC CHSL MAINS 10/01/2024 (Shift-01)

(a) 27.025 (b) 27.775

(c) 27.075 (d) 27.75

$25+2.500+0.250+0.025$

How to solve Square Root

* $\sqrt{1225}$

$\overline{12}\,\overline{25}$

35 ✓

* $\sqrt{2025}$

$\overline{20}\,\overline{25}$

45 ✓

$* \sqrt{1681}$

$\overline{16}\,\overline{81}$

41 ≈ 40 ≈ 1600 ✓

49 ≈ 50 ≈ 2500

$* \sqrt{1024}$

$\overline{10}\,\overline{24}$

32 → 900 ✓

38 → 1600

3. Find the value of the given expression.

दिए गए व्यंजक का मान ज्ञात कीजिए।

$$\sqrt{8+\sqrt{1681}}$$

SSC CGL TIER- II 02/03/2023

(a) 5 (b) 6

(c) 4 (d) 7

$= \sqrt{8+41}$

$= \sqrt{49} = 7$

$= \dfrac{4}{\sqrt{16(1+19.25)}}$

$= \dfrac{4}{\sqrt{16 \times 20.25}}$

$= \dfrac{4 \times 10}{4 \times 4.5} \; \dfrac{2}{9}$

4. Simplify (solve) the following.

निम्नलिखित को सरल (हल) कीजिए।

$$4\ \frac{\sqrt{10+\sqrt{25+\sqrt{108+\sqrt{154+\sqrt{225}}}}}}{\sqrt{16+19.25\times 4^2}}$$

SSC CGL TIER- II 06/03/2023

(a) $\frac{7}{18}$ (b) $\frac{1}{9}$

(c) $\frac{2}{9}$ (d) $\frac{5}{18}$

$\sqrt{2025} = 45$

* $\sqrt{4} = 2$

* $\sqrt{0.04} = 0.2$

* $\sqrt{0.0004} = 0.02$

* $\sqrt{0.000004} = 0.002$

$$= \frac{4}{\sqrt{16\times1+16\times19.25}}$$

$$= \frac{4}{\sqrt{16(1+19.25)}}$$

$$= \frac{4}{\sqrt{16\times20.25}}$$

$$= \frac{4\times10\ 2}{4\times4.5\ 9}$$

4. Simplify (solve) the following.

निम्नलिखित को सरल (हल) कीजिए।

$$4\ \frac{\sqrt{10+\sqrt{25+\sqrt{108+\sqrt{154+\sqrt{225}}}}}}{\sqrt{16+19.25\times4^2}}$$

SSC CGL TIER- II 06/03/2023

(a) $\frac{7}{18}$

(b) $\frac{1}{9}$

(c) $\frac{2}{9}$

(d) $\frac{5}{18}$

5. Find the value of X

X का मान ज्ञात कीजिए।

$$X = \sqrt{587 - \sqrt{121}} \div \sqrt{2316 - \sqrt{180 - \sqrt{1296}}}$$

UPSI 12/11/2021 (Shift-02)

(a) 2 (b) 1

(c) 0.5 (d) 4

$X = \sqrt{576} \div \sqrt{2316 - \sqrt{144}}$

$= 24 \div \sqrt{2316 - 12}$

$= 24 \div \sqrt{2304} = 24 \div \frac{48}{1} = 24 \times \frac{1}{48} = 0.5$

$$A = \sqrt[m]{\sqrt[n]{\sqrt[o]{\left(\left(x^{p}\right)^{q}\right)^{r}}}}$$

$$\Rightarrow A = x^{\frac{p \times q \times r}{m \times n \times o}}$$

6. If $A = \sqrt[8]{\sqrt[3]{(81^2)^3}}$ and $B = \sqrt[6]{\sqrt[3]{(3^4)^3}} \times \sqrt[6]{\sqrt[6]{(3^2)^6}}$ then what will the value A + B?

यदि $A = \sqrt[8]{\sqrt[3]{(81^2)^3}}$ तथा $B = \sqrt[6]{\sqrt[3]{(3^4)^3}} \times \sqrt[6]{\sqrt[6]{(3^2)^6}}$ तो A + B का मान क्या है?

(a) 9
(b) 4
(c) 6
(d) 3

$A = (81)^{\frac{12\times3}{4\times6\times3}} = (3)^{4\times\frac{1}{4}}$

$= 3$

$B = 3^{\frac{24\times3}{6\times3}} \times 3^{\frac{12\times6}{6\times6}}$

$= 3^{\frac{2}{3}} \times 3^{\frac{1}{3}} = 3^{\frac{2}{3}+\frac{1}{3}}$

$= 3^1$

Ans = A+B = 3+3 = 6

$a^m \times a^n = a^{m+n}$

$7^{\frac{1}{3\times3}} \times 6^{\frac{1}{3\times3}}$

$= 7 \times 6 = 42$

7. Find the value of $\left(\sqrt[3]{\sqrt[3]{343}}\right)^3 \times \left(\sqrt[3]{\sqrt[3]{216}}\right)^3$

$\left(\sqrt[3]{\sqrt[3]{343}}\right)^3 \times \left(\sqrt[3]{\sqrt[3]{216}}\right)^3$ का मान ज्ञात कीजिए।

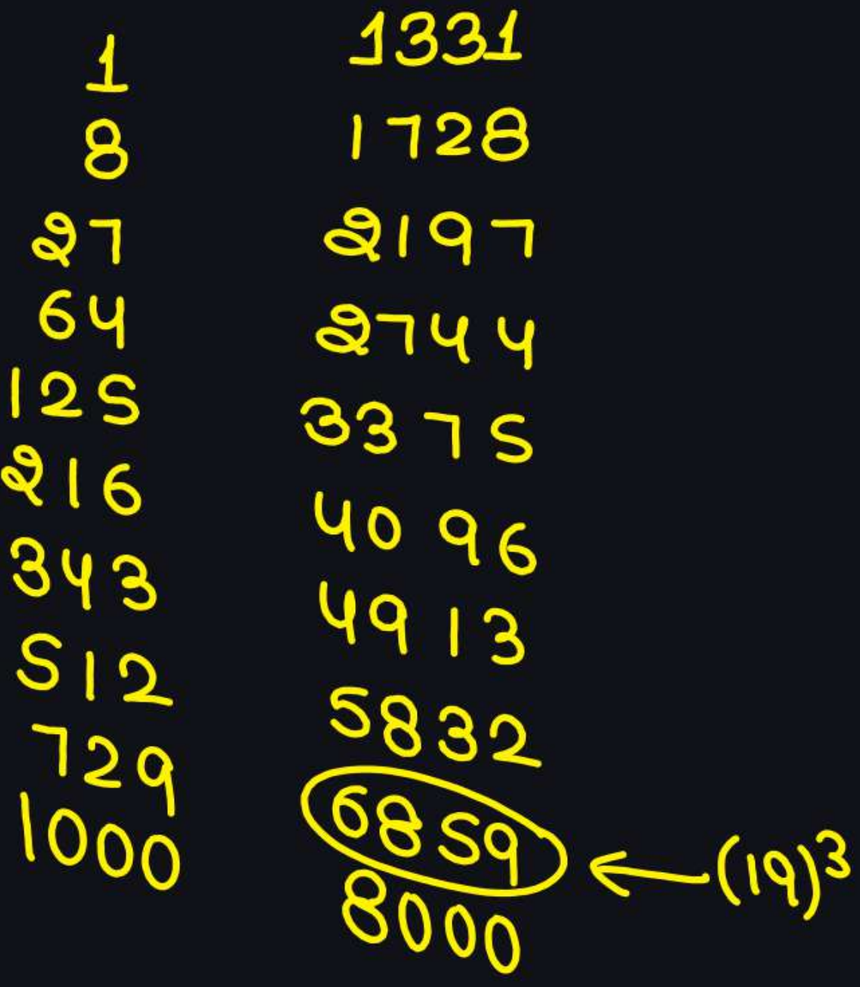
SSC CHSL MAINS 10/01/2024 (Shift-01)

(a) 42^3

(b) 56

(c) 42 ✓

(d) $\sqrt[3]{42}$


1
8
27
64
125
216
343
512
729
1000
1331
1728
2197
2744
3375
4096
4913
5832
6859
8000
$(19)^3$

8. $\dfrac{\sqrt[3]{6859}}{\sqrt[4]{1296}} \times \dfrac{3}{57} \times 42 = ?$

SSC CGL TIER-II (18/01/2025)

(a) 7
(b) 16
(c) −3
(d) 17

$$\frac{19^{3\times\frac{1}{3}}}{6^{2\times\frac{1}{4}}} \times \frac{3}{57} \times 42$$

$$= \frac{19}{6} \times \frac{3}{57} \times 42 = 7$$

9. Find the value of: $\left(\sqrt[3]{\sqrt[3]{343}}\right)^3 \times \left(\sqrt[3]{\sqrt[3]{216}}\right)^3$

Repeat of 7

$\left(\sqrt[3]{\sqrt[3]{343}}\right)^3 \times \left(\sqrt[3]{\sqrt[3]{216}}\right)^3$ का मान ज्ञात करें

SSC CHSL TIER-II 10/01/2024

(a) 42^3

(b) 56

(c) 42

(d) $\sqrt[3]{42}$

$\sqrt{100} = 10$

$\sqrt[3]{1000} = 10$

$\sqrt[3]{1000000} = 100$

10. What is the value of $\sqrt[3]{0.02 + \sqrt{0.000049}}$

का मान क्या है?

(a) 3 **(b) 0.003**

(c) 0.03 **(d) 0.3** ✓

$= \sqrt[3]{0.02 + 0.007}$

$= \sqrt[3]{0.027} = \sqrt[3]{\frac{27}{1000}} = \frac{3}{10} = 0.3$

10. What is the value of $\sqrt[3]{0.02+\sqrt{0.000049}}$

का मान क्या है?

(a) 3 **(b) 0.003**

(c) 0.03 **(d) 0.3**

$= \sqrt[3]{0.02 + 0.007}$

$= \sqrt[3]{0.027}$

$= 0.3$

$\sqrt[3]{27} = 3$ ✓

$\sqrt[3]{0.027} = 0.3$ ✓

$\sqrt[3]{0.000027} = 0.03$ ✓

* $(11)^2 = 121$
* $(111)^2 = 12321$
* $(1111)^2 = 1234321$
* $(11111)^2 = 123454321$

11. Which of the following statement(s) is/are TRUE?/निम्नलिखित में से कौन-सा/से कथन सत्य/हैहैं/?

I. $\sqrt{121} + \sqrt{12321} + \sqrt{1234321} = 1233$

II. $\sqrt{0.64} + \sqrt{64} + \sqrt{36} + \sqrt{0.36} > 15$

(a) Only I (b) Only II

(c) Neither in or II (d) Both I and II

✓ I $11 + 111 + 1111 = 1233$

✓ II $0.8 + 8 + 6 + 0.6$

$15.4 > 15$

12. $\sqrt{104.04} + \sqrt{1.0404} + \sqrt{0.010404}$ is equal to

$\sqrt{104.04} + \sqrt{1.0404} + \sqrt{0.010404}$ के बराबर है

(a) 0.306 (b) 0.0306

(c) 11.122 (d) 11.322

$10.2 + 1.02 + 0.102$

$= 11.322$

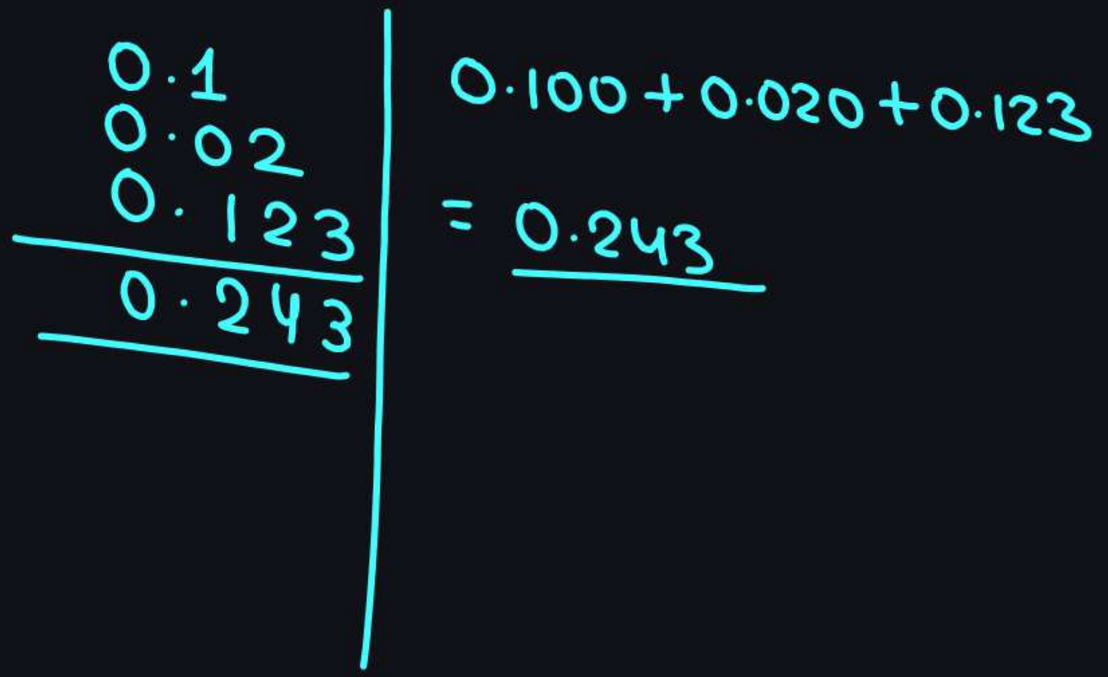
$(101)^2 = 10201$

$(102)^2 = 10404$

$(103)^2 = 10609$

$(104)^2 = 10816$

$$0.1 + 0.02 + 0.123$$



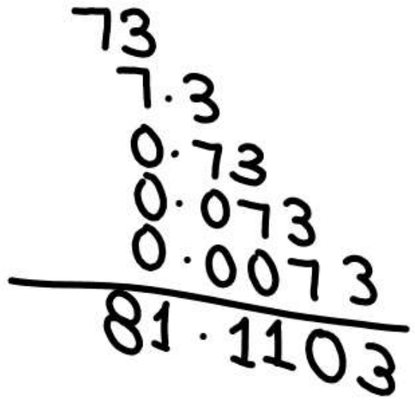
13. If $\sqrt{5329} = 73$, then value of $\sqrt{5329} + \sqrt{53.29} + \sqrt{0.5329} + \sqrt{0.005329} + \sqrt{0.00005329}$ is

यदि $\sqrt{5329} = 73$, तो $\sqrt{5329} + \sqrt{53.29} + \sqrt{0.5329} + \sqrt{0.005329} + \sqrt{0.00005329}$ का मान है

(a) 81.1003 (b) 81.0113
(c) 81.1103 (d) 81.1013

$73 + 7.3 + 0.73 + 0.073 + 0.0073$

$= 81.1103$



14. The value of $\frac{\sqrt{80}-\sqrt{112}}{\sqrt{45}-\sqrt{63}}$ is:

$$\frac{\sqrt{16\times5}-\sqrt{16\times7}}{\sqrt{9\times5}-\sqrt{9\times7}}$$

$$= \frac{\sqrt{16}\,(\sqrt{5}-\sqrt{7})}{\sqrt{9}\,(\sqrt{5}-\sqrt{7})} = \frac{4}{3} = 1\frac{1}{3}$$

$\frac{\sqrt{80}-\sqrt{112}}{\sqrt{45}-\sqrt{63}}$ का मान है:

(a) $\frac{3}{4}$

(b) $1\frac{3}{4}$

(c) $1\frac{1}{3}$

(d) $1\frac{7}{9}$

$$\sqrt{\frac{36\times2\times 3\times121\times 25\times7}{16\times2\times 49\times3\times 7\times36}}$$

$$= \frac{11\times5}{4\times7} = \frac{55}{28}$$

15. The value of $\dfrac{\sqrt{72}\times\sqrt{363}\times\sqrt{175}}{\sqrt{32}\times\sqrt{147}\times\sqrt{252}}$ is.

का मान ज्ञात करें।

(a) $\dfrac{55}{42}$

(b) $\dfrac{45}{56}$

(c) $\dfrac{45}{28}$

(d) $\dfrac{55}{28}$

$\frac{8}{100} \times \frac{8}{10} \times p = \left(\frac{8}{1000}\right)^3 \times \left(\frac{8}{10}\right)^3 \times q$

$\Rightarrow \frac{p}{q} = \frac{8^4}{10^9} = 8^4 \times 10^{-9}$

16. If $\left(\sqrt[3]{0.08 \times 0.8 \times p}\right)^3 = 0.008 \times 0.8 \times \sqrt[3]{q}\,)^3$ then find the value of $\frac{p}{q}$.

यदि $\sqrt[3]{0.08 \times 0.8 \times p} = 0.008 \times 0.8 \times \sqrt[3]{q}$, है, तो $\frac{p}{q}$ का मान ज्ञात कीजिए।

SSC CHSL MAINS 02/11/2023 (Shift- 01)

(a) $8^4 \times 10^{-8}$
(b) $8^3 \times 10^{-9}$
(c) $8^3 \times 10^{-8}$
(d) $8^4 \times 10^{-9}$

$\frac{8}{100} \times \frac{8}{10} \times p = \left(\frac{8}{10^3}\right)^3 \times \left(\frac{8}{10}\right)^3 \times q$

$\Rightarrow \frac{p}{q} = \left(\frac{8^{3+3}}{10^{9+3}}\right) \times \frac{10^3}{8^2}$

$\Rightarrow \frac{p}{q} = \frac{8^6}{8^2} \times \frac{10^3}{10^{12}}$

$\Rightarrow \frac{p}{q} = 8^4 \times 10^{-9}$

16. If $\left(\sqrt[3]{0.08 \times 0.8 \times p}\right)^3 = \frac{0.008}{1000} \times \frac{0.8}{10} \times \sqrt[3]{q}\Big)^3$ then find the value of $\frac{p}{q}$.

यदि $\sqrt[3]{0.08 \times 0.8 \times p} = 0.008 \times 0.8 \times \sqrt[3]{q}$, है, तो $\frac{p}{q}$ का मान ज्ञात कीजिए।

SSC CHSL MAINS 02/11/2023 (Shift- 01)

(a) $8^4 \times 10^{-8}$ (b) $8^3 \times 10^{-9}$

(c) $8^3 \times 10^{-8}$ (d) $8^4 \times 10^{-9}$

$a^m \times a^n = a^{m+n}$

$\frac{a^m}{a^n} = a^{m-n}$

$* \sqrt{16} = (4^2)^{\frac{1}{2}} = 4$

$* \sqrt{0.16} = 0.4$

$* \sqrt[3]{27} = (3^3)^{\frac{1}{3}} = 3$

$* \sqrt[3]{0.027} = 0.3$

$* \sqrt[4]{16} = (2^4)^{\frac{1}{4}} = 2$

$* \sqrt[4]{0.0016} = 0.2$ ✓

$A = \frac{0.02 \times 0.02}{20 \times 50 \times 3} = \frac{4}{3000} \times \frac{1}{10000}$

$= \frac{4}{3} \times \frac{1}{107}$

$B = \frac{0.9 \times 0.2}{0.4 \times 10} = \frac{18}{40}$

$ans = \frac{4}{3} \times \frac{1}{107} \times \frac{18}{40}\ 6$

$= \frac{6}{108} = 6 \times 10^{-8}$

17. If $A = \frac{\sqrt{0.0004} \times \sqrt[3]{0.000008}}{\sqrt[4]{16000} \times \sqrt[3]{125000} \times \sqrt[4]{810}}$ and

$B = \frac{\sqrt[3]{0.729} \times \sqrt[4]{0.0016}}{\sqrt{0.16}}$, then what is $A \times B$?

यदि $A = \frac{\sqrt{0.0004} \times \sqrt[3]{0.000008}}{\sqrt[4]{16000} \times \sqrt[3]{125000} \times \sqrt[4]{810}}$ और $B = \frac{\sqrt[3]{0.729} \times \sqrt[4]{0.0016}}{\sqrt{0.16}}$ है, तो $A \times B$ क्या है?

$\sqrt[4]{160000} \times \sqrt[3]{125000} \times \sqrt[4]{81}$

SSC CGL MAINS (08/08/2022)

(a) 5×10^{-8}

(b) 7×10^{-7}

(c) $\left(\frac{7}{4}\right) \times 10^{-8}$

(d) 6×10^{-8}

18. What is the value of

$$\frac{\sqrt{29.16}}{\sqrt{1.1664}}+\frac{\sqrt{0.2916}}{\sqrt{116.64}}+\frac{\sqrt{0.0036}}{\sqrt{0.36}}\ ?$$

$$\frac{\sqrt{29,16}}{\sqrt{1,1664}}+\frac{\sqrt{0,2916}}{\sqrt{116,64}}+\frac{\sqrt{0,0036}}{\sqrt{0,36}}$$ का मान क्या है?

SSC CGL MAINS (08/08/2022

(a) $\frac{101}{20}$

(b) $\frac{103}{20}$

(c) $\frac{26}{5}$

(d) $\frac{27}{5}$

$$\sqrt{\frac{2916}{11664}\times\frac{10000}{100}}+\sqrt{\frac{2916}{11664}\times\frac{100}{10000}}+\sqrt{\frac{36}{36}\times\frac{100}{10000}}$$

$$=\frac{10}{2}+\frac{1}{20}+\frac{1}{10}$$

$$=\frac{100+1+2}{20}=\frac{103}{20}$$

$$\sqrt{\frac{1.24}{2.64}\times\frac{2.79}{5.94}} = \sqrt{\frac{31}{66}\times\frac{31}{66}}$$

$$= \frac{31}{66}$$

19. Find the value of $\dfrac{\sqrt{1.24}\times\sqrt{2.79}}{\sqrt{2.64}\times\sqrt{5.94}}$

$\dfrac{\sqrt{1.24}\times\sqrt{2.79}}{\sqrt{2.64}\times\sqrt{5.94}}$ का मान ज्ञात कीजिए।

SSC CGL TIER II 26/10/2023

(a) $\dfrac{31}{44}$

(b) $\dfrac{33}{64}$

(c) $\dfrac{31}{66}$ ✓

(d) $\dfrac{33}{31}$

TYPE – 02

SMALLEST & GREATEST VALUE

Type-1

$2^3 < 2^4$

Type-2

$2^3 < 3^3$

$8 < 27$

M-1

$5^{\frac{1}{2}}$ $7^{\frac{1}{3}}$ $13^{\frac{1}{4}}$

LCM (2,3,4) = 12

$5^{\frac{1}{2}\times 12^{6}}$ $7^{\frac{1}{3}\times 12^{4}}$ $13^{\frac{1}{4}\times 12^{3}}$

5^6 7^4 13^3

15625 > 2401 > 2197

20. Which is the largest among the numbers, $\sqrt{5}$, $\sqrt[3]{7}$, $\sqrt[4]{13}$

जो संख्याओं $\sqrt{5}$, $\sqrt[3]{7}$, $\sqrt[4]{13}$ में सबसे बड़ी है

(a) $\sqrt{5}$ ✓

(b) $\sqrt[3]{7}$

(c) $\sqrt[4]{13}$

(d) All are equal

$5^{\frac{1}{2}\times 63}$ $7^{\frac{1}{3}\times 62}$

$125 > 49$

$5^{\frac{1}{2}\times 4\times 2}$ $13^{\frac{1}{4}\times 4}$

$25 > 13$

20. Which is the largest among the numbers,

$\sqrt{5}$, $\sqrt[3]{7}$, $\sqrt[4]{13}$

जो संख्याओं $\sqrt{5}$, $\sqrt[3]{7}$, $\sqrt[4]{13}$ में सबसे बड़ी है

(a) $\sqrt{5}$

(b) $\sqrt[3]{7}$

(c) $\sqrt[4]{13}$

(d) All are equal

$6^{\frac{1}{3}\times 6^2}$ $5^{\frac{1}{2}\times 6^3}$ $12^{\frac{1}{6}\times 6}$

36 125 12

21. **Which one among $\sqrt[3]{6}, \sqrt[2]{5}$ and $\sqrt[6]{12}$ is the largest?**

$\sqrt[3]{6}, \sqrt[2]{5}$ और $\sqrt[6]{12}$ में से कौन सबसे बड़ा है?

(a) $\sqrt[3]{6}$ (b) $\sqrt[2]{5}$

(c) $\sqrt[6]{12}$ (d) All are equal

$\sqrt{a} = \sqrt[2]{a} = a^{\frac{1}{2}}$

$4^{\frac{1}{3}\times 12}$ $2^{\frac{1}{2}\times 12}$ $3^{\frac{1}{6}\times 12}$ $5^{\frac{1}{4}\times 12}$

4^4 2^6 3^2 5^3

256 64 9 125

$256 > 125 > 64 > 9$

$\sqrt[3]{4} > \sqrt[4]{5} > \sqrt{2} > \sqrt[6]{3}$

22. Arranging the following in descending order, we get $\sqrt[3]{4}, \sqrt{2}, \sqrt[6]{3}, \sqrt[4]{5}$

निम्नलिखित को अवरोही क्रम में व्यवस्थित करने पर, हमें $\sqrt[3]{4}, \sqrt{2}, \sqrt[6]{3}, \sqrt[4]{5}$ प्राप्त होता है

(a) $\sqrt[3]{4} > \sqrt[4]{5} > \sqrt{2} > \sqrt[6]{3}$

(b) $\sqrt[4]{5} > \sqrt[3]{4} > \sqrt[6]{3} > \sqrt{2}$

(c) $\sqrt{2} > \sqrt[6]{3} > \sqrt[3]{4} > \sqrt[4]{5}$

(d) $\sqrt[6]{3} > \sqrt[4]{5} > \sqrt[3]{4} > \sqrt{2}$

0.3 0.4 0.5 0.6

23. The greatest among the numbers $\sqrt{0.09}$, $\sqrt[3]{0.064}$, 0.5 and $\frac{3}{5}$ is

$\sqrt{0.09}$, $\sqrt[3]{0.064}$, 0.5 और $\frac{3}{5}$ में से सबसे बड़ी संख्या है

(a) $\sqrt{0.09}$ (b) $\frac{3}{5}$

(c) 0.5 (d) $\sqrt[3]{0.064}$

SSC Maths- Chapter-20

Page-348 (Eng)
363 (Hindi)

3^{34} 2^{51} 7^{17}

HCF(34, 51, 17) = 17

$(3^2)^{17}$ $(2^3)^{17}$ $(7^1)^{17}$

$(9)^{17}$ $(8)^{17}$ $(7)^{17}$

Ascending $(7)^{17} < (8)^{17} < (9)^{17}$

$7^{17} < 2^{51} < 3^{34}$

24. Arranging the following in ascending order 3^{34}, 2^{51}, 7^{17} we get

निम्नलिखित को आरोही क्रम में व्यवस्थित करना 3^{34}, 2^{51}, 7^{17} हम प्राप्त करते हैं

(a) $3^{34} > 2^{51} > 7^{17}$

(b) $7^{17} < 2^{51} < 3^{34}$

(c) $3^{34} > 7^{17} > 2^{51}$

(d) $2^{51} > 3^{34} > 7^{17}$

25. The smallest among the numbers 2^{250}, 3^{150}, 5^{100} and 4^{200}

2^{250}, 3^{150}, 5^{100} और 4^{200} संख्याओं में सबसे छोटी

(a) 4^{200} **(b) 5^{100}**

(c) 3^{150} **(d) 2^{250}**

(2^5), (3^3), (5^2), (4^4)

32, 27, 25, 256

26. Which of the following is true?

निम्नलिखित में से कौन सा सही है?

(a) $(22.5)^{27} > (7.5)^{54}$ (b) $(22.5)^{27} < (7.5)^{54}$

(c) $(22.3)^{27} = (7.5)^{54}$ (d) $(22.3)^{27} \geq (7.5)^{54}$

(a) $(22.5)^{27}$ $(7.5^2)^{27}$

$\Rightarrow (22.5)^{27} < (56.25)^{27}$

Same Sum

$(\sqrt{8}+\sqrt{3}) < (\sqrt{6}+\sqrt{5})$

Trick

gap $5 < 1$

gap ↓ value ↑

Same Difference

$(\sqrt{11}-\sqrt{8}) < (\sqrt{8}-\sqrt{5})$

digit ↓ value ↑

Same Product

$(\sqrt{8} \pm \sqrt{3})$ $(\sqrt{6} \pm \sqrt{4})$

Sum $11 > 10$

Diff $5 > 2$

Sum/Diff ↑ Value ↑

$\sqrt{6}+\sqrt{2} < \sqrt{5}+\sqrt{3}$

gap 4 2

gap ↓ value ↑

27. Choose the incorrect reaction(s) from the following:

निम्नलिखित में से गलत प्रतिक्रिया चुनें:

(i) $\sqrt{6}+\sqrt{2}=\sqrt{5}+\sqrt{3}$

(ii) $\sqrt{6}+\sqrt{2}<\sqrt{5}+\sqrt{3}$

(iii) $\sqrt{6}+\sqrt{2}>\sqrt{5}+\sqrt{3}$

(a) (ii) and (iii) **(b) (i)**

(c) (i) and (ii) **(d) (i) and (iii)**

28. Which value among $\sqrt{11}+\sqrt{5}$, $\sqrt{14}+\sqrt{2}$, $\sqrt{8}+\sqrt{8}$, is the largest?

gap 6 12 0

$\sqrt{11}+\sqrt{5}$, $\sqrt{14}+\sqrt{2}$, $\sqrt{8}+\sqrt{8}$ में से कौन सा मान सबसे बड़ा है:

(a) $\sqrt{11}+\sqrt{5}$ (b) $\sqrt{14}+\sqrt{2}$

(c) $\sqrt{8}+\sqrt{8}$ (d) All are equal

gap ↓ value ↑

29. Which of the following statements(s) is/are true?

निम्नलिखित में से कौन सा/से कथन सत्य है/हैं?

I. $\sqrt{11}+\sqrt{7} < \sqrt{10}+\sqrt{8}$

II. $\sqrt{17}+\sqrt{11} > \sqrt{15}+\sqrt{13}$

(a) Only I
(b) Only II
(c) Both I and II
(d) Neither I nor II

Digit ↓ value ↑
Digit ↑ value ↓

30. Arranging $a = \sqrt{6} - \sqrt{5}, b = \sqrt{5} - \sqrt{4}, c = \sqrt{4} - \sqrt{3}$ in ascending order, we get.

$a = \sqrt{6} - \sqrt{5}, b = \sqrt{5} - \sqrt{4}, c = \sqrt{4} - \sqrt{3}$ को आरोही क्रम में रखने पर, हम पाते हैं।

(a) $c < b < a$ **(b) $b < a < c$**

(c) $a < c < b$ **(d) $a < b < c$** ✓

$(\sqrt{6} - \sqrt{5}) < (\sqrt{5} - \sqrt{4} < \sqrt{4} - \sqrt{3}$

$a < b < c$

31. The greatest among $\sqrt{7}-\sqrt{5}, \sqrt{5}-\sqrt{3}, \sqrt{9}-\sqrt{7}, \sqrt{11}-\sqrt{9}$ is

$\sqrt{7}-\sqrt{5}, \sqrt{5}-\sqrt{3}, \sqrt{9}-\sqrt{7}, \sqrt{11}-\sqrt{9}$ में सबसे बड़ा है

(a) $\sqrt{7}-\sqrt{5}$ (b) $\sqrt{5}-\sqrt{3}$

(c) $\sqrt{9}-\sqrt{7}$ (d) $\sqrt{11}-\sqrt{9}$

32. Which is the greatest among $(\sqrt{19}-\sqrt{17})$, $(\sqrt{13}-\sqrt{11})$, $(\sqrt{7}-\sqrt{5})$ and $(\sqrt{5}-\sqrt{3})$?

$(\sqrt{19}-\sqrt{17})$, $(\sqrt{13}-\sqrt{11})$, $(\sqrt{7}-\sqrt{5})$ और $(\sqrt{5}-\sqrt{3})$ में सबसे बड़ा कौन है?

(a) $\sqrt{19}-\sqrt{17}$ (b) $\sqrt{13}-\sqrt{11}$

(c) $\sqrt{7}-\sqrt{5}$ (d) $\sqrt{5}-\sqrt{3}$

33. Which is the greatest among ?

$\sqrt{6}+\sqrt{4}$, $\sqrt{8}+\sqrt{3}$, $\sqrt{12}+\sqrt{2}$ and $\sqrt{24}+\sqrt{1}$

10 11 14 25

$\sqrt{6}+\sqrt{4}$, $\sqrt{8}+\sqrt{3}$, $\sqrt{12}+\sqrt{2}$ और $\sqrt{24}+\sqrt{1}$ में सबसे बड़ा कौन है?

(a) $\sqrt{6}+\sqrt{4}$

(b) $\sqrt{8}+\sqrt{3}$

(c) $\sqrt{12}+\sqrt{2}$

(d) $\sqrt{24}+\sqrt{1}$

34. Which is the smallest among ?

$\sqrt{5} - \sqrt{4}$, $\sqrt{10} - \sqrt{2}$, $\sqrt{20} - \sqrt{1}$

$\sqrt{5}^{1} - \sqrt{4}$, $\sqrt{10}^{8} - \sqrt{2}$, $\sqrt{20}^{19} - \sqrt{1}$ में सबसे छोटा कौन है?

(a) $\sqrt{5} - \sqrt{4}$

(b) $\sqrt{10} - \sqrt{2}$

(c) $\sqrt{20} - \sqrt{1}$

(d) All are equal

* $(2^3)^2$

$= 2^{3 \times 2}$

$= 2^6$

* 2^{3^2}

$= 2^9$

35. Which of the following is greatest?

निम्न में से कौन सबसे बड़ा है।

(a) $3^{2^3} = 3^8$

(b) $3^{3^2} = 3^9$

(c) $2^{3^2} = 2^9$

(d) $(2^3)^2 = 2^6$

TYPE – 03

INFINITE SERIES

* $\sqrt{a\sqrt{a\sqrt{a\ldots\infty}}}$

ans = a

* $\sqrt{a\sqrt{a\sqrt{a}}}$

ans = $a^{\frac{2^n-1}{2^n}}$

36. Find the value of $\sqrt{a\sqrt{a\sqrt{a.....\infty}}}$

$\sqrt{a\sqrt{a\sqrt{a.....\infty}}}$ का मान ज्ञात कीजिए।

(a) a
(b) $a^{\frac{3}{2}}$
(c) a^3
(d) 0

37. $\sqrt{3\sqrt{3\sqrt{3.....}}}$ is equal to.

$\sqrt{3\sqrt{3\sqrt{3.....}}}$ के बराबर है:

(a) $\sqrt{3}$
(b) 3
(c) $2\sqrt{3}$
(d) $3\sqrt{3}$

38. If $x = \sqrt{4\sqrt{4\sqrt{4\sqrt{4...}}}}$, then what is the value of x?

यदि $x = \sqrt{4\sqrt{4\sqrt{4\sqrt{4...}}}}$ है, तो x का मान क्या है?

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(a) 2
(b) 4
(c) 8
(d) 16

39. $\sqrt{7\sqrt{7\sqrt{7\sqrt{7....\infty}}}} = 343^{y-1}$ then y = ?

$\sqrt{7\sqrt{7\sqrt{7\sqrt{7....\infty}}}} = 343^{y-1}$ तो y = ?

(a) $\frac{4}{3}$ (b) $\frac{3}{2}$

(c) $\frac{5}{1}$ (d) 1

$7 = (7^3)^{y-1}$

$\Rightarrow 7^1 = 7^{3y-3}$

$\Rightarrow 1 = 3y-3$

$\Rightarrow 4 = 3y$

$\Rightarrow \frac{4}{3} = y$

Q. $y = \sqrt{a\sqrt{a\sqrt{a\ldots\infty}}}$

Ans $y^2 = a\underbrace{\sqrt{a\sqrt{a\sqrt{a\ldots\infty}}}}_{y}$

$\Rightarrow y^2 = a \times y$

$\therefore \boxed{y = a}$

---

* $y = \sqrt{7\sqrt{7\sqrt{7\ldots\infty}}}$

ans $\boxed{y = 7}$

$y=\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4}\ldots}}}\infty$

$\Rightarrow y^2=2\sqrt[3]{4\sqrt{2\sqrt[3]{4}\ldots}}\infty$

$\Rightarrow y^2=2\sqrt[3]{4\times y}$

$\Rightarrow y^6=8\times 4\times y^1$

$\Rightarrow y^5=32$

$\Rightarrow y^5=2^5$

$\therefore y=2$

40. The value of $\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4}\ldots.}}}}}$ is

$\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4}\ldots.}}}}}$ का मान है

(a) 2
(b) 2^2
(c) 2^3
(d) 2^5

$y=\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4}\ldots\infty}}}$

$\Rightarrow y^2 = 2\sqrt[3]{4\sqrt{2\sqrt[3]{4}\ldots\infty}}$

$\Rightarrow (y^2)^3 = (2)^3 \times 4 \times y$

$\Rightarrow y^6 = 8 \times 4 \times y^1$

$\Rightarrow y^5 = 2^5$

$y = 2$

40. **The value of** $\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4}\ldots.}}}}}$ **is**

$\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4}\ldots.}}}}}$ का मान है

(a) 2 (b) 2^2

(c) 2^3 (d) 2^5

41. $\sqrt{27 \div \sqrt{27 \div \sqrt{27 \div} \ldots\ldots \infty} = ?}$

$\sqrt{27 \div \sqrt{27 \div \sqrt{27 \div} \ldots\ldots \infty} = ?}$

(a) $\sqrt[3]{27}$ (b) 27

(c) $\sqrt[2]{27}$ (d) $\sqrt[4]{27}$

$y = \sqrt{27 \div \sqrt{27 \div \sqrt{27} \ldots \infty}}$

$\therefore \quad y^2 = 27 \div \sqrt{27 \div \sqrt{27} \ldots \infty}$

$\Rightarrow \quad y^2 = 27 \div y$

$\Rightarrow \quad y^2 = \frac{27}{y}$

$\Rightarrow \quad y^3 = 27$

$\Rightarrow \quad y = \sqrt[3]{27}$

$$\sqrt{x}^{\sqrt{x}^{\sqrt{x}^{\cdots\cdots\infty}}} = \frac{1}{2}$$

$$\Rightarrow (\sqrt{x})^{\frac{1}{x}\times 2} = \left(\frac{1}{2}\right)^{1\times 2}$$

$$\Rightarrow \sqrt{x} = \frac{1}{4}$$

$$\Rightarrow x = \frac{1}{16}$$

42. If $\sqrt{x}^{\sqrt{x}^{\sqrt{x}^{\cdots\cdots\cdots\infty}}} = \frac{1}{2}$ then x = ?

यदि $\sqrt{x}^{\sqrt{x}^{\sqrt{x}^{\cdots\cdots\cdots\infty}}} = \frac{1}{2}$ तो x = ?

(a) $\frac{1}{8}$

(b) $\frac{1}{4}$

(c) $\frac{1}{16}$

(d) $\frac{1}{32}$

Infinite Terms

Q. $x=\sqrt{a\sqrt{a\sqrt{a}\ldots}}\ \infty$

Ans $x=a$

Finite Terms

Q. $x=\sqrt{a\sqrt{a\sqrt{a}}}$

$2^3=8$

$ans = a^{\frac{7}{8}}$

Q $x=\sqrt{a\sqrt{a\sqrt{a\sqrt{a}}}}$

$2^4=16$

$ans = a^{\frac{15}{16}}$

$2^4 = 16$

ans = $a^{\frac{15}{16}}$

43. Find the value of $\sqrt{a\sqrt{a\sqrt{a\sqrt{a}}}}$

$\sqrt{a\sqrt{a\sqrt{a\sqrt{a}}}}$ का मान ज्ञात कीजिए-

(a) $a^{\frac{7}{8}}$

(b) $a^{\frac{15}{16}}$

(c) a

(d) a^4

$2^6 = 64$

ans = $7^{\frac{63}{64}}$

44. Find the value of $\sqrt{7\sqrt{7\sqrt{7\sqrt{7\sqrt{7\sqrt{7}}}}}}$

$\sqrt{7\sqrt{7\sqrt{7\sqrt{7\sqrt{7\sqrt{7}}}}}}$ का मान ज्ञात कीजिए-

(a) $7^{\frac{63}{64}}$

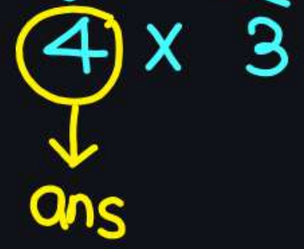
(b) $7^{\frac{31}{32}}$

(c) $7^{\frac{15}{16}}$

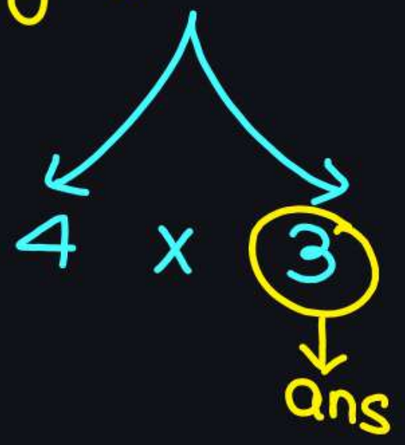
(d) 7^7

* $y=\sqrt{12+\sqrt{12+\sqrt{12+\ldots\infty}}}$

Tricic



* $y=\sqrt{12-\sqrt{12-\sqrt{12-\ldots\infty}}}$



45. $\sqrt{12+\sqrt{12+\sqrt{12+.....}}}$ **is equal to.**

$\sqrt{12+\sqrt{12+\sqrt{12+.....}}}$ के बराबर है:

$y=\sqrt{12+\sqrt{12+\sqrt{12+..}}}\ \infty$

4 x 3

(a) 3
(b) 4
(c) 6
(d) 2

46. Find the value of

$$\sqrt{30+\sqrt{30+\sqrt{30+\sqrt{30+.....\infty}}}}$$

6×5

$$\sqrt{30+\sqrt{30+\sqrt{30+\sqrt{30+.....\infty}}}}$$

का मान ज्ञात कीजिए

(a) 3 (b) 4

(c) 5 (d) 6

47. Let $x = \sqrt{272 + \sqrt{272 + \sqrt{272 + \sqrt{272 + ... + \infty}}}}$ then x equals

माना $x = \sqrt{272 + \sqrt{272 + \sqrt{272 + \sqrt{272 + ... + \infty}}}}$ तो x बराबर है

$272 = 17 \times 16$

(a) 16
(b) $4\sqrt{13}$
(c) 17
(d) 4.35

48. **Find the value of the given expression.**

दिए गए व्यंजक का मान ज्ञात कीजिए।

$$\sqrt{20-\sqrt{20-\sqrt{20-\sqrt{20-\ldots\ldots\infty}}}}$$

5 x 4

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(a) 4

(b) 6

(c) 5

(d) 2

8 ÷ 4

= 2

49. What is the value of :

$$\left(\sqrt{72-\sqrt{72-\sqrt{72-....\infty}}}\right) \div$$

9 x 8 ✓

$$\left(\sqrt{20-\sqrt{20-\sqrt{20-....\infty}}}\right)$$ का मान क्या है?

5 x 4 ✓

(a) 4 (b) 2

(c) 3.6 (d) 8

50. Let $x = \sqrt{42 - \sqrt{42 - \sqrt{42 - \sqrt{42 - \ldots - \infty}}}}$ then x equals

7x6

माना $x = \sqrt{42 - \sqrt{42 - \sqrt{42 - \sqrt{42 - \ldots - \infty}}}}$ तो x बराबर है

(a) 6 (b) 7

(c) Between 6 and 7 (d) Greater than 7

Q. $x = \sqrt{a+\sqrt{a+\sqrt{a..}}}\infty$

ans $x = \dfrac{\sqrt{1+4a}+1}{2}$

Q. $x = \sqrt{a-\sqrt{a-\sqrt{a..}}}\infty$

$x = \dfrac{\sqrt{1+4a}-1}{2}$

51. $\sqrt{5+\sqrt{5+\sqrt{5+....\infty}}}$ =?

ans = $\frac{\sqrt{1+4a}+1}{2}$

= $\frac{\sqrt{21}+1}{2}$

(a) $\frac{\sqrt{21}+1}{2}$

(b) $\frac{\sqrt{21}-1}{2}$

(c) $\frac{\sqrt{21}+3}{2}$

(d) $\frac{\sqrt{21}-3}{2}$

52. $\sqrt{31+\sqrt{31+\sqrt{31+\sqrt{31+....\infty}}}}$ =?

(a) $5\sqrt{5}-1.5$

(b) $2.5\sqrt{5}+0.5$

(c) $\frac{5\sqrt{5}-1}{2}$

(d) $\frac{2\sqrt{31}+1}{2}$

ans $= \frac{\sqrt{1+4a}+1}{2}$

$= \frac{\sqrt{125}+1}{2}$

$= \frac{5\sqrt{5}+1}{2}$

$= 2.5\sqrt{5}+0.5$

53. $\sqrt{1-\sqrt{1-\sqrt{1-....\infty}}} = ?$

(a) $\frac{\sqrt{5}+1}{2}$

(b) $\frac{\sqrt{5}-1}{2}$

(c) $\frac{\sqrt{5}+3}{2}$

(d) $\frac{\sqrt{5}-3}{2}$

ans $= \frac{\sqrt{1+4a}-1}{2}$

$= \frac{\sqrt{5}-1}{2}$

* $x = \sqrt{35 \pm 1\sqrt{35 \pm \sqrt{35 \pm \ldots}}}\infty$

a b

$$x = \frac{\sqrt{1+4a} \pm 1}{2}$$

* $x = \sqrt{35 \pm 2\sqrt{35 \pm 2\sqrt{35 \pm \ldots}}}\infty$

a b

$$x = \frac{\sqrt{b^2+4a} \pm b}{2}$$

① $x = \sqrt{a + b\sqrt{a + b\sqrt{a}\,..\,\infty}}$

$$x = \frac{\sqrt{b^2+4a} + b}{2}$$

$$x - y = \frac{\sqrt{b^2+4a} + b - \sqrt{b^2+4a} + b}{2}$$

$$= \frac{2b}{2}$$

$$= b$$

* $y = \sqrt{a - b\sqrt{a - b\sqrt{a}\,..\,\infty}}$

$$y = \frac{\sqrt{b^2+4a} - b}{2}$$

$$xxy = \left(\frac{\sqrt{b^2+4a} + b}{2}\right)\left(\frac{\sqrt{b^2+4a} - b}{2}\right)$$

$$= \frac{(\sqrt{b^2+4a})^2 - (b)^2}{4}$$

$$= \frac{b^2 + 4a - b^2}{4} = a$$

Q. $x = \sqrt{35 + 2\sqrt{35 + 2\sqrt{35 \ldots}}}\infty$

(a ↓ , b ↓)

find i) $x - y = ?$

ii) $x \times y = ?$

Q. $y = \sqrt{35 - 2\sqrt{35 - 2\sqrt{35 \ldots}}}\infty$

Ans

$a - b$

i) $x - y = (b) = 2$

ii) $x \times y = (a) = 35$

$a \times b$

Q. $x = \sqrt{35 + 2\sqrt{35 + 2\sqrt{35 \ldots \infty}}}$

(a, b)

Q. $y = \sqrt{35 - 2\sqrt{35 - 2\sqrt{35 \ldots \infty}}}$

$x = ?$ $\quad y = ?$

Ans

$$\sqrt{35 \pm 2\sqrt{35 \pm 2\sqrt{35 \ldots \infty}}}$$

7 × 5

\+ −

Q. $x = \sqrt{15+2\sqrt{15+2\sqrt{15..\infty}}}$ Q $y = \sqrt{15-2\sqrt{15-2\sqrt{15..\infty}}}$

Ans

$$x = \frac{\sqrt{b^2+4a}+b}{2}$$

$$= \frac{\sqrt{4+60}+2}{2}$$

$$= 5$$

$$y = \frac{\sqrt{b^2+4a}-b}{2}$$

$$= \frac{\sqrt{4+60}-2}{2}$$

$$= 3$$

0. $x = \sqrt{15 + 2\sqrt{15 + 2\sqrt{15 \ldots \infty}}}$



Ans

$x = 5$

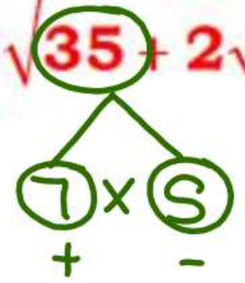
$y = 3$

0 $y = \sqrt{15 - 2\sqrt{15 - 2\sqrt{15 \ldots \infty}}}$

54. **Find** $\sqrt{35+2\sqrt{35+2\sqrt{35+2\sqrt{35+....\infty}}}}$ =?

$\sqrt{35+2\sqrt{35+2\sqrt{35+2\sqrt{35+....\infty}}}}$ ज्ञात करें

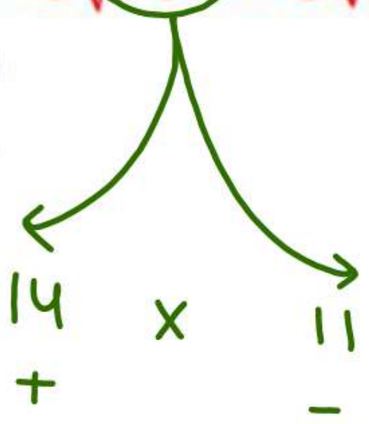
(a) 6
(b) 7
(c) 5
(d) 6.4



55. Find $\sqrt{154 - 3\sqrt{154 - 3\sqrt{154 - 3\sqrt{154 - \ldots.\infty}}}}$ =?

$\sqrt{154 - 3\sqrt{154 - 3\sqrt{154 - 3\sqrt{154 - \ldots.\infty}}}}$ ज्ञात करें?

(a) 13
(b) 14
(c) 11
(d) 9



$$(a+b)^2 = (a-b)^2 + 4ab$$

or

$$(a-b)^2 = (a+b)^2 - 4ab$$

56. If $P = \sqrt{11 + 3\sqrt{11 + 3\sqrt{11 + 3\sqrt{11 - ....\infty}}}}$ and $Q = \sqrt{11 - 3\sqrt{11 - 3\sqrt{11 - 3\sqrt{11 - ....\infty}}}}$ then P + Q = ?

यदि $P = \sqrt{11 + 3\sqrt{11 + 3\sqrt{11 + 3\sqrt{11 - ....\infty}}}}$ और $Q = \sqrt{11 - 3\sqrt{11 - 3\sqrt{11 - 3\sqrt{11 - ....\infty}}}}$ तो P + Q = ?

(a) $\sqrt{47}$ (b) $\sqrt{65}$

(c) $\sqrt{41}$ (d) $\sqrt{53}$

$P - Q = 3$

$P \times Q = 11$

$(P+Q)^2 = (P-Q)^2 + 4PQ$

$\Rightarrow (P+Q)^2 = 9 + 44$

$\therefore (P+Q) = \sqrt{53}$

Q. $P=\sqrt{8+3\sqrt{8+3\sqrt{8}\ldots\infty}}$ $\quad Q=\sqrt{8-3\sqrt{8-3\sqrt{8}\ldots\infty}}$

$P+Q=?$

Ans

$$(P+Q)^2=(P-Q)^2+4PQ$$

$$\Rightarrow (P+Q)^2=(3)^2+4\times 8$$

$$\Rightarrow (P+Q)=\sqrt{41}$$

$$* \sqrt{a + b\sqrt{a - b\sqrt{a + b\sqrt{a \ldots}}}}\infty = \frac{\sqrt{4a - 3b^2} + b}{2}$$

$$* \sqrt{a - b\sqrt{a + b\sqrt{a - b\sqrt{a \ldots}}}}\infty = \frac{\sqrt{4a - 3b^2} - b}{2}$$

57. $\sqrt{7+2\sqrt{7-2\sqrt{7+2\sqrt{7-2\sqrt{7+\ldots.}}}}}$ =?

(a) $\sqrt{51}$ (b) 4

(c) $\frac{3+\sqrt{15}}{2}$ (d) 3

ans = $\frac{\sqrt{4a-3b^2}+b}{2}$

= $\frac{\sqrt{28-12}+2}{2}$

= $\frac{4+2}{2} = 3$

$$x = \frac{\sqrt{4a - 3b^2} - b}{2}$$

$$= \frac{\sqrt{24 - 3} - 1}{2}$$

$$= \frac{\sqrt{21} - 1}{2}$$

58. Let $x = \sqrt{6 - 1\sqrt{6 + \sqrt{6 - \sqrt{6 + \ldots.}}}}$ to infinity ; then x equals

a b

माना $x = \sqrt{6 - \sqrt{6 + \sqrt{6 - \sqrt{6 + \ldots.}}}}$ से अनंत तक; तो x बराबर है

(a) 3

(b) $\sqrt{21}$

(c) $\frac{\sqrt{21} - 1}{2}$

(d) $\frac{\sqrt{21} + 1}{2}$

Type	Result
* $x=\sqrt{a\pm\sqrt{a\pm\sqrt{a}\ldots\infty}}$	$=\frac{\sqrt{4a+1}\pm 1}{2}$
* $x=\sqrt{a\pm b\sqrt{a\pm b\sqrt{a}\ldots\infty}}$	$=\frac{\sqrt{4a+b^2}\pm b}{2}$
* $x=\sqrt{a+b\sqrt{a-b\sqrt{a}\ldots\infty}}$	$=\frac{\sqrt{4a-3b^2}+b}{2}$
or	
* $x=\sqrt{a-b\sqrt{a+b\sqrt{a}\ldots\infty}}$	$=\frac{\sqrt{4a-3b^2}-b}{2}$

TYPE – 04

IF SQUARE ROOT IS INSIDE SQUARE ROOT

Case 1:

Q. $x = \sqrt{8+2\sqrt{15}}$

Ans $x = \sqrt{5} + \sqrt{3}$

Q $y = \sqrt{11+2\sqrt{18}}$

Ans $y = \sqrt{9} + \sqrt{2}$

$= 3 + \sqrt{2}$

Case 2:

Q $y = \sqrt{5+\sqrt{24}}$

Ans $y = \sqrt{\frac{10+2\sqrt{24}}{2}}$

$\Rightarrow y = \frac{\sqrt{6}+\sqrt{4}}{\sqrt{2}}$

$y = \sqrt{3}+\sqrt{2}$

Q $y = \sqrt{8+\sqrt{63}}$

Ans $y = \sqrt{\frac{8+\sqrt{63}}{1}}$

$\Rightarrow y = \sqrt{\frac{16+2\sqrt{63}}{2}}$

$\Rightarrow y = \frac{\sqrt{9}+\sqrt{7}}{\sqrt{2}}$

$y = \frac{3+\sqrt{7}}{\sqrt{2}}$

Case 3:

Q. $y=\sqrt{16+6\sqrt{7}}$

Ans $y=\sqrt{16+2\times3\sqrt{7}}$

$=\sqrt{16+2\sqrt{3^2\times7}}$

$=\sqrt{16+2\sqrt{63}}$

$=\sqrt{9}+\sqrt{7}$

$=3+\sqrt{7}$ ✓

Q. $y=\sqrt{11+6\sqrt{2}}$

Ans $y=\sqrt{11+2\times3\sqrt{2}}$

$=\sqrt{11+2\sqrt{18}}$

$=\sqrt{9}+\sqrt{2}$

$=$ $3+\sqrt{2}$ ✓

59. If $x = \left(\sqrt{31 + 2\sqrt{30}}\right) - \left(\sqrt{31 - 2\sqrt{30}}\right)$, then what is the value of x?

यदि $x = \left(\sqrt{31+2\sqrt{30}}\right) - \left(\sqrt{31-2\sqrt{30}}\right)$ तो x का मान क्या है?

CRPF HCM 22/02/2023 (Shift - 02)

(a) 2

(b) $2\sqrt{2}$

(c) $2\sqrt{15}$

(d) 4

$x = (\sqrt{30} + \sqrt{1}) - (\sqrt{30} - \sqrt{1})$

$= \sqrt{30} + 1 - \sqrt{30} + 1$

$= 2$

1×30

$$y = \sqrt{14 + 6\sqrt{5}}$$

$$= \sqrt{14 + 2 \times 3\sqrt{5}}$$

$$= \sqrt{14 + 2\sqrt{45}}$$

$$= \sqrt{9} + \sqrt{5}$$

$$= 3 + \sqrt{5}$$

60. The square root of $14 + 6\sqrt{5}$

$14 + 6\sqrt{5}$ का वर्गमूल है

(a) $2 + \sqrt{5}$

(b) $3 + \sqrt{5}$

(c) $5 + \sqrt{3}$

(d) $3 + 2\sqrt{5}$

61. **If the positive square root of $41 + 24\sqrt{2}$ is A and positive square root of $36 - 16\sqrt{2}$ is B, then what is the value of A – B?**

यदि $41 + 24\sqrt{2}$ का धनात्मक वर्गमूल **A** और $36 - 16\sqrt{2}$ का धनात्मक वर्गमूल **B** है, तो **A – B** का मान क्या है?

(a) 3 (b) 1

(c) 2 (d) 5

$A - B = \left(\sqrt{41+24\sqrt{2}}\right) - \left(\sqrt{36-16\sqrt{2}}\right)$

$= \left(\sqrt{41+2\times12\sqrt{2}}\right) - \left(\sqrt{36-2\times8\sqrt{2}}\right)$

$= \left(\sqrt{41+2\sqrt{288}}\right) - \left(\sqrt{36-2\sqrt{128}}\right)$

$= \left(\sqrt{32}+\sqrt{9}\right) - \left(\sqrt{32}-\sqrt{4}\right)$

$= \sqrt{32} + 3 - \sqrt{32} + 2$

$= 5$

4×32

62. If $\sqrt{52 - 30\sqrt{3}} = a + b\sqrt{3}$, then $(a + b)$ is equal to:

यदि $\sqrt{52 - 30\sqrt{3}} = a + b\sqrt{3}$, तो $(a + b)$ बराबर है:

(a) 4 (b) –4

(c) –2 (d) 2

$\sqrt{52 - 2 \times 15\sqrt{3}}$

$= \sqrt{52 - 2\sqrt{675}}$

$= \sqrt{27} - \sqrt{25}$

$\Rightarrow 3\sqrt{3} - 5 = a + b\sqrt{3}$

$3\sqrt{3} = b\sqrt{3}$ | $a = -5$

$b = 3$

$a + b = -5 + 3$

$= -2$

63. If $\sqrt{54 - 20\sqrt{2}} = a + b\sqrt{2}$, then $(2a + 3b)$ is equal to:

यदि $\sqrt{54 - 20\sqrt{2}} = a + b\sqrt{2}$, तो $(2a + 3b)$ बराबर है:

(a) 8 (b) 4

(c) −3 (d) 11

$\sqrt{54 - 2\sqrt{200}}$

$= \sqrt{50} - \sqrt{4}$

$\Rightarrow 5\sqrt{2} - 2 = a + b\sqrt{2}$

$a = -2$ | $5\sqrt{2} = b\sqrt{2}$

$b = 5$

$2a + 3b$

$= -4 + 15$

$= 11$

64. If $x = \dfrac{5\sqrt{3} - \sqrt{48} - 4\sqrt{2} + \sqrt{50}}{\sqrt{3}\times\sqrt{3} + \sqrt{6}}$ and $y = \sqrt{7 - 4\sqrt{3}}$ then the value of $(x - y)$ is closest to:

यदि $x = \dfrac{5\sqrt{3} - \sqrt{48} - 4\sqrt{2} + \sqrt{50}}{3 + \sqrt{6}}$ और $y = \sqrt{7 - 4\sqrt{3}}$ तो $(x - y)$ का मान निकटतम है:

(a) 0.3 (b) 0.8
(c) 0.4 (d) 0.6

$\sqrt{3} = 1.732$

$$x = \frac{5\sqrt{3} - 4\sqrt{3} - 4\sqrt{2} + 5\sqrt{2}}{\sqrt{3}(\sqrt{3} + \sqrt{2})}$$

$$= \frac{(\sqrt{3} + \sqrt{2})}{\sqrt{3}(\sqrt{3} + \sqrt{2})} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

$$y = \sqrt{7 - 4\sqrt{3}}$$

$$= \sqrt{7 - 2\sqrt{12}} = \sqrt{4} - \sqrt{3} = (2 - \sqrt{3})$$

$$x - y = \frac{\sqrt{3}}{3} - (2 - \sqrt{3})$$

$$= \frac{\sqrt{3} - 6 + 3\sqrt{3}}{3} = \frac{4\sqrt{3} - 6}{3} = \frac{4 \times 1.73 - 6}{3} = \frac{6.92 - 6}{3} = \frac{0.92}{3} = 0.30$$

65. If $x = \sqrt{-\sqrt{3} + \sqrt{3 + 8\sqrt{7 + 4\sqrt{3}}}}$ where $x > 0$, then the value of x is equal to:

यदि $x = \sqrt{-\sqrt{3} + \sqrt{3 + 8\sqrt{7 + 4\sqrt{3}}}}$ जहाँ $x > 0$, तो x का मान इसके बराबर है:

(a) 3 (b) 4

(c) 1 (d) 2

$$x = \sqrt{-\sqrt{3} + \sqrt{3 + 8\sqrt{7 + 2\sqrt{12}}}}$$

$$= \sqrt{-\sqrt{3} + \sqrt{3 + 8(\sqrt{4} + \sqrt{3})}}$$

$$= \sqrt{-\sqrt{3} + \sqrt{3 + 8(2 + \sqrt{3})}}$$

$$= \sqrt{-\sqrt{3} + \sqrt{19 + 8\sqrt{3}}}$$

$$= \sqrt{-\sqrt{3} + \sqrt{19 + 2\sqrt{48}}}$$

$$= \sqrt{-\sqrt{3} \sqrt{16} + \sqrt{3}}$$

$$= \sqrt{4} = 2$$

$$* (a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ac)$$

66. Evaluate $\sqrt{10+2\sqrt{6}+2\sqrt{10}+2\sqrt{15}}$

मूल्यांकन करे $\sqrt{10+2\sqrt{6}+2\sqrt{10}+2\sqrt{15}}$

(a) $\sqrt{2}+\sqrt{3}$ (b) $\sqrt{3}+\sqrt{5}$

(c) $(\sqrt{2}+\sqrt{3}+\sqrt{5})$ (d) $\sqrt{3}+\sqrt{5}+\sqrt{7}$

$$= \sqrt{2+3+5+2(\sqrt{6}+\sqrt{10}+\sqrt{15})}$$

$$= \sqrt{(\sqrt{2})^2+(\sqrt{3})^2+(\sqrt{5})^2+2(\sqrt{6}+\sqrt{15}+\sqrt{10})}$$

$$= \sqrt{(\sqrt{2}+\sqrt{3}+\sqrt{5})^2}$$

$$= (\sqrt{2}+\sqrt{3}+\sqrt{5})$$

TYPE – 05

IF SQUARE ROOT IS IN THE DENOMINATOR

$$\sqrt{\frac{(\sqrt{3}+\sqrt{2})}{(\sqrt{3}-\sqrt{2})} \times \frac{(\sqrt{3}+\sqrt{2})}{(\sqrt{3}+\sqrt{2})}}$$

$$= \sqrt{\frac{(\sqrt{3}+\sqrt{2})^2}{(3-2)}}$$

$$= \frac{(\sqrt{3}+\sqrt{2})}{1}$$

67. **The square root of $\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$ is**

$\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$ का वर्गमूल है

(a) $\sqrt{3}+\sqrt{2}$ (b) $\sqrt{3}-\sqrt{2}$

(c) $2\sqrt{2}+\sqrt{3}$ (d) $\sqrt{2}-\sqrt{3}$

* $\frac{1}{\sqrt{7}-\sqrt{6}} = \frac{\sqrt{7}+\sqrt{6}}{1}$

* $\frac{1}{\sqrt{8}+\sqrt{7}} = \frac{\sqrt{8}-\sqrt{7}}{1}$

* $\frac{1}{\sqrt{5}-\sqrt{3}} = \frac{\sqrt{5}+\sqrt{3}}{2}$

67. The square root of $\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$ is

$$= \sqrt{\frac{(\sqrt{3}+\sqrt{2})}{(\sqrt{3}-\sqrt{2})}}$$

$$= \sqrt{(\sqrt{3}+\sqrt{2}) \times \frac{(\sqrt{3}+\sqrt{2})}{1}}$$

$$= (\sqrt{3}+\sqrt{2})$$

$\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$ का वर्गमूल है

(a) $\sqrt{3}+\sqrt{2}$ (b) $\sqrt{3}-\sqrt{2}$

(c) $2\sqrt{2}+\sqrt{3}$ (d) $\sqrt{2}-\sqrt{3}$

68. $2+\frac{6}{\sqrt{3}}+\frac{1}{2+\sqrt{3}}+\frac{1}{\sqrt{3}-2}$ **equals to**

$2+\frac{6}{\sqrt{3}}+\frac{1}{2+\sqrt{3}}+\frac{1}{\sqrt{3}-2}$ बराबर है

(a) $(2-\sqrt{3})$ (b) $-(2+\sqrt{3})$

(c) 1 (d) 2

Ans : $2+\frac{2\sqrt{3}}{\sqrt{3}\times\sqrt{3}}+2-\sqrt{3}+\frac{(\sqrt{3}+2)}{-1}$

$= 2+2\sqrt{3}+2-\sqrt{3}-\sqrt{3}-2$

$= 2$

69. $\frac{3\sqrt{2}}{\sqrt{6}+\sqrt{3}} - \frac{2\sqrt{6}}{\sqrt{3}+1} + \frac{2\sqrt{3}}{\sqrt{6}+2}$ is equal to

$\frac{3\sqrt{2}}{\sqrt{6}+\sqrt{3}} - \frac{2\sqrt{6}}{\sqrt{3}+1} + \frac{2\sqrt{3}}{\sqrt{6}+2}$ के बराबर है

(a) 3 (b) 2

(c) 0 (d) $\sqrt{3}$

$$\frac{3\sqrt{2}(\sqrt{6}-\sqrt{3})}{3} - \frac{2\sqrt{6}(\sqrt{3}-1)}{2} + \frac{2\sqrt{3}(\sqrt{6}-2)}{2}$$

$$= \sqrt{12} - \sqrt{6} - \sqrt{18} + \sqrt{6} + \sqrt{18} - 2\sqrt{3}$$

$$= 2\sqrt{3} - 2\sqrt{3}$$

$$= 0$$

$\sqrt{2} \times \sqrt{6} = \sqrt{2 \times 6}$

$= \sqrt{12}$

70. The value of $\frac{1}{\sqrt{2}+1}+\frac{1}{\sqrt{3}+\sqrt{2}}+\frac{1}{\sqrt{4}+\sqrt{3}}+....+\frac{1}{\sqrt{100}+\sqrt{99}}$ is

$\frac{1}{\sqrt{2}+1}+\frac{1}{\sqrt{3}+\sqrt{2}}+\frac{1}{\sqrt{4}+\sqrt{3}}+....+\frac{1}{\sqrt{100}+\sqrt{99}}$ का मान है

(a) 1 (b) 9

(c) $\sqrt{99}$ (d) $\sqrt{99}-1$

$\frac{(\sqrt{2}-1)}{1}+\frac{\sqrt{3}-\sqrt{2}}{1}+\frac{\sqrt{4}-\sqrt{3}}{1}+.....+\frac{\sqrt{100}-\sqrt{99}}{1}$

$= \sqrt{100}-1$

$= 10-1$

$= 9$

71. **What is the value of**

$$\frac{1}{\sqrt{3}+\sqrt{4}}+\frac{1}{\sqrt{4}+\sqrt{5}}+\frac{1}{\sqrt{5}+\sqrt{6}}+\frac{1}{\sqrt{6}+\sqrt{7}}\ ?$$

$$\frac{1}{\sqrt{3}+\sqrt{4}}+\frac{1}{\sqrt{4}+\sqrt{5}}+\frac{1}{\sqrt{5}+\sqrt{6}}+\frac{1}{\sqrt{6}+\sqrt{7}}$$ का मान क्या है?

(a) $\sqrt{3}-\sqrt{7}$ (b) $\sqrt{6}-\sqrt{5}$

(c) $\sqrt{7}-\sqrt{3}$ (d) $\sqrt{8}-\sqrt{5}$

$$\frac{1}{\sqrt{4}+\sqrt{3}}+\frac{1}{\sqrt{5}+\sqrt{4}}+\frac{1}{\sqrt{6}+\sqrt{5}}+\frac{1}{\sqrt{7}+\sqrt{6}}$$

$$=(\sqrt{4}-\sqrt{3})+\sqrt{5}-\sqrt{4}+\sqrt{6}-\sqrt{5}+(\sqrt{7}-\sqrt{6})$$

$$=\sqrt{7}-\sqrt{3}$$

72. The value of $\frac{1}{\sqrt{3.25}+\sqrt{2.25}} + \frac{1}{\sqrt{4.25}+\sqrt{3.25}}$

$+ \frac{1}{\sqrt{5.25}+\sqrt{4.25}} + \frac{1}{\sqrt{6.25}+\sqrt{5.25}}$ is

का मान है

(a) 1.00 (b) 1.25

(c) 1.50 (d) 2.25

$= (\sqrt{3.25} - \sqrt{2.25}) + (\sqrt{4.25} - \sqrt{3.25}) + (\sqrt{5.25} - \sqrt{4.25}) + (\sqrt{6.25} - \sqrt{5.25})$

$= \sqrt{6.25} - \sqrt{2.25}$

$= 2.5 - 1.5$

$= 1$

73. Simplify $\frac{1}{\sqrt{100}-\sqrt{99}} - \frac{1}{\sqrt{99}-\sqrt{98}} +$

$\frac{1}{\sqrt{98}-\sqrt{97}} - \frac{1}{\sqrt{97}-\sqrt{96}} + \ldots + \frac{1}{\sqrt{2}-\sqrt{1}}$

सरल कीजिए।

(a) 0 (b) 9

(c) 10 (d) 11

$+ (\sqrt{100}+\sqrt{99}) - (\sqrt{99}+\sqrt{98}) + \ldots\ldots + \sqrt{2}+1$

$= (\sqrt{100}+\sqrt{99}) - \sqrt{99} - \sqrt{98} + \ldots\ldots - (\sqrt{2}+1)$

$= \sqrt{100} + 1$

$= 10 + 1$

$= 11$

$(4+\sqrt{15})-(\sqrt{15}+\sqrt{14})+\ldots\ldots-(3+\sqrt{8})$

$= 4-\sqrt{8}$

$= 4-2\sqrt{2}$

74. The value of

$$\frac{1}{4-\sqrt{15}}-\frac{1}{\sqrt{15}-\sqrt{14}}+\frac{1}{\sqrt{14}-\sqrt{13}}-\frac{1}{\sqrt{13}-\sqrt{12}}+$$

$$\frac{1}{\sqrt{12}-\sqrt{11}}-\frac{1}{\sqrt{11}-\sqrt{10}}+\frac{1}{\sqrt{10}-3}-\frac{1}{3-\sqrt{8}}$$ **is:**

$$\frac{1}{4-\sqrt{15}}-\frac{1}{\sqrt{15}-\sqrt{14}}+\frac{1}{\sqrt{14}-\sqrt{13}}-\frac{1}{\sqrt{13}-\sqrt{12}}+$$

$$\frac{1}{\sqrt{12}-\sqrt{11}}-\frac{1}{\sqrt{11}-\sqrt{10}}+\frac{1}{\sqrt{10}-3}-\frac{1}{3-\sqrt{8}}$$ का मान ज्ञात करें।

SSC CGL MAINS 29/01/2022

(a) $2+2\sqrt{2}$

(b) $2-2\sqrt{2}$

(c) $4-2\sqrt{2}$

(d) $4+2\sqrt{2}$

75. What is the value of $\frac{\sqrt{7}+\sqrt{5}}{\sqrt{7}-\sqrt{5}} \div \frac{\sqrt{14}+\sqrt{10}}{\sqrt{14}-\sqrt{10}} + \frac{\sqrt{10}}{\sqrt{5}}$?

$\frac{\sqrt{7}+\sqrt{5}}{\sqrt{7}-\sqrt{5}} \div \frac{\sqrt{14}+\sqrt{10}}{\sqrt{14}-\sqrt{10}} + \frac{\sqrt{10}}{\sqrt{5}}$ का मान क्या है?

SSC CGL MAINS (08/08/2022)

(a) $\sqrt{2}+1$ ✓

(b) $2\sqrt{2}+2$

(c) $\sqrt{2}+2$

(d) $2\sqrt{2}+1$

$$\frac{\sqrt{7}+\sqrt{5}}{\sqrt{7}-\sqrt{5}} \times \frac{\sqrt{14}-\sqrt{10}}{\sqrt{14}+\sqrt{10}} + \sqrt{\frac{10}{5}}^2$$

$$= \frac{(\sqrt{7}+\sqrt{5})}{(\sqrt{7}-\sqrt{5})} \times \frac{\sqrt{2}(\sqrt{7}-\sqrt{5})}{\sqrt{2}(\sqrt{7}+\sqrt{5})} + \sqrt{2}$$

$$= 1+\sqrt{2}$$

76. The value of

$$\frac{1}{\sqrt{(12-\sqrt{140})}}-\frac{1}{\sqrt{(8-\sqrt{60})}}-\frac{2}{\sqrt{10+\sqrt{84}}}$$

$$\frac{\sqrt{12+\sqrt{140}}}{\sqrt{4}}-\frac{\sqrt{(8+\sqrt{60})}}{\sqrt{4}}-\frac{2\sqrt{10-\sqrt{84}}}{\sqrt{16}}$$

$$=\frac{\sqrt{12+2\sqrt{35}}}{2}-\frac{\sqrt{8+2\sqrt{15}}}{2}-\frac{2\sqrt{10-2\sqrt{21}}}{4\,2}$$

$$=\frac{\sqrt{7}+\sqrt{5}-\sqrt{5}-\sqrt{3}-\sqrt{7}+\sqrt{3}}{2}$$

$$=0$$

$$\frac{1}{\sqrt{(12-\sqrt{140})}}-\frac{1}{\sqrt{(8-\sqrt{60})}}-\frac{2}{\sqrt{10+\sqrt{84}}}$$ का मान है

(a) 0
(b) 1
(c) 2
(d) 3

77. If $\frac{\sqrt{26-7\sqrt{3}}}{\sqrt{14+5\sqrt{3}}} = \frac{b+a\sqrt{3}}{11}$ b > 0, then what is the value of $\sqrt{(b-a)}$?

यदि $\frac{\sqrt{26-7\sqrt{3}}}{\sqrt{14+5\sqrt{3}}} = \frac{b+a\sqrt{3}}{11}$ b > 0 हो, तो $\sqrt{(b-a)}$ का मान कितना होगा?

SSC CGL MAINS 03/02/2022

(a) 5 (b) 25
(c) 12 (d) 9

$$\frac{\sqrt{26-\sqrt{147}}}{\sqrt{14+\sqrt{75}}}$$

$$= \frac{\sqrt{52-2\sqrt{147}}}{\sqrt{28+2\sqrt{75}}} = \frac{\sqrt{49}-\sqrt{3}}{\sqrt{25}+\sqrt{3}}$$

$$= \frac{7-\sqrt{3}}{(5+\sqrt{3})} \times \frac{(5-\sqrt{3})}{(5-\sqrt{3})}$$

$$= \frac{(7-\sqrt{3})(5-\sqrt{3})}{22} = \frac{35-7\sqrt{3}-5\sqrt{3}+3}{22} = \frac{38-12\sqrt{3}}{22} = \frac{19-6\sqrt{3}}{11} = \frac{b+a\sqrt{3}}{11}$$

$a=-6$, $b=19$

$\sqrt{19-(-6)} = 5$

$$= \frac{\sqrt{38-\sqrt{75}}}{\sqrt{26+\sqrt{147}}} = \frac{\sqrt{76-2\sqrt{75}}}{\sqrt{52+2\sqrt{147}}}$$

$$= \frac{\sqrt{75}-\sqrt{1}}{\sqrt{49}+\sqrt{3}} = \frac{(5\sqrt{3}-1)\times(7-\sqrt{3})}{(7+\sqrt{3})(7-\sqrt{3})}$$

$$= \frac{35\sqrt{3}-15-7+\sqrt{3}}{49-3}$$

$$= \frac{36\sqrt{3}-22}{46} = \frac{18\sqrt{3}-11}{23} = \frac{a+b\sqrt{3}}{23}$$

$18\sqrt{3} = b\sqrt{3}$

$b = 18$

$a = -11$

$b - a$
$= 18-(-11)$
$= 29$

78. $\frac{\sqrt{38-5\sqrt{3}}}{\sqrt{26+7\sqrt{3}}} = \frac{a+b\sqrt{3}}{23}$, **b > 0, then the value of (b – a) is:**

यदि $\frac{\sqrt{38-5\sqrt{3}}}{\sqrt{26+7\sqrt{3}}} = \frac{a+b\sqrt{3}}{23}$, **b > 0** है, तो **(b – a)** का मान कितना होगा?

SSC CGL MAINS 03/02/2022

(a) 7
(b) 18
(c) 29 ✓
(d) 11

79. ** The value of $5\sqrt{3}+7\sqrt{2}-\sqrt{6}-\frac{23}{\sqrt{2}+\sqrt{3}+\sqrt{6}}$ is:

$5\sqrt{3}+7\sqrt{2}-\sqrt{6}-\frac{23}{\sqrt{2}+\sqrt{3}+\sqrt{6}}$ is: का मान ज्ञात कीजिए।

(a) 15
(b) 16
(c) 12
(d) 10

Ans = $5\times1.732+7\times1.414-2.440-\frac{23}{5.586}$

$= 8.660+9.898-2.440-\frac{23000}{5586}$

$= 16.118-4.110$

≈ 12.008

$\sqrt{2} = 1.414$
$\sqrt{3} = 1.732$
$\sqrt{6} = 2.440$

Sum = 5.586

ans= $\frac{1}{0.91} - \frac{1}{2.55}$

$= \frac{100}{91} - \frac{100}{255} = \frac{25500-9100}{91 \times 255}$

$= \frac{16400}{23205} =$

80. $\left[\frac{1}{\sqrt{2}+\sqrt{3}-\sqrt{5}} + \frac{1}{\sqrt{2}-\sqrt{3}-\sqrt{5}}\right]$ **in simplified form equals**

सरलीकृत रूप से $\left[\frac{1}{\sqrt{2}+\sqrt{3}-\sqrt{5}} + \frac{1}{\sqrt{2}-\sqrt{3}-\sqrt{5}}\right]$ के बराबर है

~~(a) 1~~

~~(b) $\sqrt{2} = 1.414$~~

(c) $\frac{1}{\sqrt{2}} = \frac{1}{1.414}$

~~(d) 0~~

$\sqrt{2} = 1.414$
$\sqrt{3} = 1.732$
$\sqrt{5} = 2.236$

0.910

$$\frac{\sqrt{2}-\sqrt{3}-\sqrt{5}+\sqrt{2}+\sqrt{3}-\sqrt{5}}{(\sqrt{2}+\sqrt{3}-\sqrt{5})(\sqrt{2}-\sqrt{3}-\sqrt{5})}$$

$$= \frac{2\sqrt{2}-2\sqrt{5}}{2-\sqrt{6}-\sqrt{10}+\sqrt{6}-3-\sqrt{15}-\sqrt{10}+\sqrt{15}+5}$$

$$= \frac{2(\sqrt{2}-\sqrt{5})}{4-2\sqrt{10}} = \frac{2(\sqrt{2}-\sqrt{5})}{2(2-\sqrt{10})}$$

$$= \frac{1(\sqrt{2}-\sqrt{5})}{\sqrt{2}(\sqrt{2}-\sqrt{5})} = \frac{1}{\sqrt{2}}$$

80. $\left[\frac{1}{\sqrt{2}+\sqrt{3}-\sqrt{5}}+\frac{1}{\sqrt{2}-\sqrt{3}-\sqrt{5}}\right]$ **in simplified form equals**

सरलीकृत रूप से $\left[\frac{1}{(\sqrt{2}+\sqrt{3}-\sqrt{5})}+\frac{1}{(\sqrt{2}-\sqrt{3}-\sqrt{5})}\right]$ के बराबर है

(a) 1

(b) $\sqrt{2}$

(c) $\frac{1}{\sqrt{2}}$

(d) 0

TYPE – 06

SIMPLIFICATION OF POWER

* $\left((a^m)^n\right)^o = a^{m \times n \times o}$

* $\sqrt[r]{\sqrt[q]{\sqrt[p]{a}}} = a^{\frac{1}{p \times q \times r}}$

* $\sqrt[r]{\sqrt[q]{\sqrt[p]{\left((a^m)^n\right)^o}}} = a^{\frac{m \times n \times o}{p \times q \times r}}$

$5^{\frac{2}{\cancel{9\times4}\ \cancel{6\times3}}} \times 5^{\frac{2}{\cancel{9\times4}\ \cancel{6\times3}}}$

$= 5^2 \times 5^2$

$= 5^4$

81. Simplify: $\left[\sqrt[3]{\sqrt[6]{5^9}}\right]^4 \left[\sqrt[3]{\sqrt[6]{5^9}}\right]^4$

सरल करे: $\left[\sqrt[3]{\sqrt[6]{5^9}}\right]^4 \left[\sqrt[3]{\sqrt[6]{5^9}}\right]^4$

(a) 5^2 **(b)** 5^4

(c) 5^8 **(d)** 5^{12}

$a^m \times a^n = a^{m+n}$

$$\frac{32^{0.20}}{2^{0.25} \times (2^2)^{0.075} \times (2^3)^{0.2}}$$

$$= \frac{(2^5)^{0.20}}{2^{0.25+0.150+0.6}}$$

$$= \frac{2^{1.00}}{2^{1.000}} = 1$$

82. **Solve:** $\dfrac{\left(32^{0.13} \times 32^{0.07}\right)}{\left(2^{0.25} \times 4^{0.075} \times 8^{0.2}\right)}$

हल करे $\dfrac{\left(32^{0.13} \times 32^{0.07}\right)}{\left(2^{0.25} \times 4^{0.075} \times 8^{0.2}\right)}$

(a) 2 (b) $\frac{1}{3}$

(c) 1 (d) $\frac{1}{2}$

83. **The value of**

$$\frac{(625)^{6.25} \times (25)^{2.6}}{(5^4)^{6.75} \times (\sqrt{5})^{2.4}} \times \frac{(0.49)^4 \times (0.343)^4}{(0.2401)^5}$$ **is:**

$$\frac{(625)^{6.25} \times (25)^{2.6}}{(5^4)^{6.75} \times (\sqrt{5})^{2.4}} \times \frac{(0.49)^4 \times (0.343)^4}{(0.2401)^5}$$ का मान है:

TGT SST (Female) 14/11/2018 (Shift- 01)

(a) 25 (b) 0.35

(c) 175 (d) 0.25

$$\frac{5^{4\times6.25} \times 5^{2\times2.6}}{5^{4\times6.75} \times 5^{\frac{2.4}{2}}} \times \frac{(0.7)^{2\times4} \times (0.7)^{3\times4}}{(0.7)^{4\times5}}$$

$$= \frac{5^{25+5.2}}{5^{27+1.2}} \times \frac{0.7^{8+12}}{0.7^{20}}$$

$$= \frac{5^{30.2}}{5^{28.2}} \times \frac{0.7^{20}}{0.7^{20}}$$

$$= 5^2 \times 1$$

$$= 25$$

$a^m \times a^n = a^{m+n}$

$\frac{a^m}{a^n} = a^{m-n}$

84. On simplification

$$\frac{(49)^{\frac{1}{2}} \times (512)^{0.25} \times (8)^{0.25} \times (729)^{\frac{2}{3}} \div (64)^{\frac{4}{3}}}{(343)^{\frac{1}{3}} \times (256)^{-0.375} \times (81)^{0.5}}$$

reduces to / का न्यूनतमक मान है-

PGT CS (Female) 26/07/2018 (Shift- 01)

(a) 2.25 (b) 1.25

(c) 1.5 (d) 2.5

$$\frac{7^{2\times\frac{1}{2}} \times 2^{9\times\frac{1}{4}} \times 2^{3\times\frac{1}{4}} \times 9^{3\times\frac{2}{3}} \div 2^{6\times\frac{4}{3}}}{7^{3\times\frac{1}{3}} \times 2^{8\times(-\frac{3}{8})} \times 3^{24\times\frac{1}{12}}}$$

$$= \frac{7^1 \times 2^{\frac{12}{4}} \times 81 \times 1}{7 \times 2^{-3} \times 9 \times 2^8}$$

$$= \frac{8\times 9}{2^5} = \frac{8\times 9}{32} = 2.25$$

यदि $\frac{3^{a+3} \times 4^{a+6} \times 25^{a+1}}{27^{a-1} \times 8^{a-2} \times 125^{a+4}} = \frac{4}{15^{26}}$ है तो $\sqrt{a+9}$ का मान है:

(a) 4 (b) 6

(c) ~~5~~ (d) 8

$$\frac{3^{a+3} \times 2^{2(a+6)} \times 5^{2(a+1)}}{3^{3(a-1)} \times 2^{3(a-2)} \times 5^{3(a+4)}} = \frac{2^2}{3^{26} \times 5^{26}}$$

$$\Rightarrow \frac{2^{-a+18}}{3^{2a-6} \times 5^{a+10}} = \frac{2^2}{3^{26} \times 5^{26}}$$

$-a+18 = 2$

$16 = a$

$2a-6 = 26$

~~$2a = 32$~~ $a = 16$

$a+10 = 26$

$a = 16$

$\sqrt{16+9}$

$= 5$

यदि $\frac{3^{a+3} \times 4^{a+6} \times 25^{a+1}}{27^{a-1} \times 8^{a-2} \times 125^{a+4}} = \frac{4}{15^{26}}$ है तो $\sqrt{a+9}$ का मान है:

(a) 4 (b) 6

(c) 5 (d) 8

$$\frac{2^{2(a+6)}}{2^{3(a-2)}} = 2^2$$

$$\Rightarrow 2^{-a+18} = 2^2$$

$$-a+18 = 2$$

$$\Rightarrow 16 = a$$

$$\sqrt{16+9} = 5$$

$$\frac{3^{\frac{5}{5}\times n} \times 3^{2n+1}}{3^{2n} \times 3^{n-1}}$$

$$= 3^{n+2n+1-2n-n+1}$$

$$= 3^2 = 9$$

86. The value of $\frac{(243)^{\frac{n}{5}} \times 3^{2n+1}}{9^n \times 3^{n-1}}$ is:

$\frac{(243)^{\frac{n}{5}} \times 3^{2n+1}}{9^n \times 3^{n-1}}$ का मान है:

(a) 3
(b) 9
(c) 6
(d) 12

$$\frac{\cancel{(2+3)}}{\cancel{(2+3)}} \quad \frac{(2+3)}{(4^1+6^1)} \quad \frac{(2)^2+(3)^2}{(4)^2+(6)^2} \quad \frac{(2)^3+(3)^3}{(4)^3+(6)^3}$$

$$= 1 \qquad = \frac{\cancel{(2+3)}}{2\,\cancel{(2+3)}} \qquad = \frac{1}{2^2} \qquad = \frac{1}{2^3}$$

$$= \frac{1}{2^1} \qquad = \frac{1}{4} \qquad = \frac{1}{8}$$

87. Simplify the following expression

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$$8\left(\frac{0.2\times0.2\times0.2+0.04\times0.04\times0.04}{0.4\times0.4\times0.4+0.08\times0.08\times0.08}\right)+9$$

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(a) 17 (b) 10

(c) 1 (d) 8

$$\frac{8\left[(0.2)^3+(0.04)^3\right]}{\left[(0.4)^3+(0.08)^3\right]}+9$$

$$=\frac{8}{}\times\frac{1}{2^3}+9$$

$$=10$$

88. If $27^{2x-1} = (243)^3$ then value of x is

यदि $27^{2x-1} = (243)^3$ तो x का मान है।

(a) 3 (b) 6

(c) 7 (d) 9

$3^{3(2x-1)} = 3^{5\times3}$

$\Rightarrow 3(2x-1) = 5\times3$

$\Rightarrow 2x = 6$

$x = 3$

89. If $3^{x+y} = 81$ and $81^{x-y} = 3$, then the value of x is
यदि $3^{x+y} = 81$ और $81^{x-y} = 3$, तो x का मान है

(a) 42 (b) $\frac{15}{8}$

(c) $\frac{17}{8}$ (d) 39

$3^{x+y} = 3^4$ $\quad$ $3^{4(x-y)} = 3^1$

$x+y = 4$ $\quad$ $4(x-y) = 1$

$x-y = \frac{1}{4}$

$2x = 4 + \frac{1}{4} = \frac{17}{4}$

$\therefore x = \frac{17}{8}$

$2x-y = 1.5$

$x+y = 1.5$

$3x = 3$

$x = 1$

$y = 0.5$

Ans $= 3^{0.5}$

$= 3^{\frac{1}{2}}$

$= \sqrt{3}$

90. **If $3^{2x-y} = 3^{x+y} = \sqrt{27}$, then the value of 3^{x-y} will be:**

यदि $3^{2x-y} = 3^{x+y} = \sqrt{27}$, तब 3^{x-y} का मान होगा:

(a) 3

(b) $\frac{1}{\sqrt{3}}$

(c) $\sqrt{3}$

(d) $\frac{1}{\sqrt{27}}$

$3^{2x-y} = 3^{x+y} = 3^{\frac{3}{2}}$

$2x-y = x+y = \frac{3}{2}$

91. If $2^{x-1} + 2^{x+1} = 320$, then the value of x is

यदि $2^{x-1} + 2^{x+1} = 320$, तो x का मान है

(a) 0 (b) 9

(c) 10 (d) 7

ⓐ $2^{-1} + 2^{1} = \frac{1}{2} + 2 = \frac{5}{2}$

ⓓ $2^{6} + 2^{8} = 64 + 256 = 320$

$2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 16$, $2^5 = 32$

$2^6 = 64$, $2^7 = 128$, $2^8 = 256$, $2^9 = 512$, $2^{10} = 1024$

92. If $2^x + 3^y = 17$ and $2^{x+2} - 3^{y+1} = 5$, then:

यदि $2^x + 3^y = 17$ तथा $2^{x+2} - 3^{y+1} = 5$ है, तो:

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(a) $x = 1, y = 3$
(b) $x = 3, y = 3$
(c) $x = 3, y = 2$
(d) $x = 1, y = 2$

(a) $2^1 + 3^3 = 29$

(b) $2^3 + 3^3 =$

(c) $2^3 + 3^2 = 17$ ✓

$2^5 - 3^3 = 5$ ✓

93. If $5^x - 3^y = 13438$ and $5^{x-1} + 3^{y+1} = 9686$, then $x+y = ?$

यदि $5^x - 3^y = 13438$ और $5^{x-1} + 3^{y+1} = 9686$, तो $x+y = ?$

(a) 9
(b) 11
(c) 13
(d) 15

$5^6 - 3^7$
$= 15625 - 2187$
$= 13438$

$x+y$
$= 6+7$
$= 13$

$5^1 = 5$
$5^2 = 25$
$5^3 = 125$
$5^4 = 625$
$5^5 = 3125$
$5^6 = 15625$

$3^1 = 3$
$3^2 = 9$
$3^3 = 27$
$3^4 = 81$
$3^5 = 243$
$3^6 = 729$
$3^7 = 2187$

94. If $x^{x\sqrt{x}} = (x\sqrt{x})^x$ then, x equals

यदि $x^{x\sqrt{x}} = (x\sqrt{x})^x$ है, तो x बराबर है

(a) $\frac{4}{9}$ (b) $\frac{2}{3}$

(c) $\frac{9}{4}$ (d) $\frac{3}{2}$

$x^{x\sqrt{x}} = x^{\frac{3}{2}x}$

$\Rightarrow x\sqrt{x} = \frac{3}{2}x$

$\Rightarrow x = \frac{9}{4}$

$(x \cdot x \cdot x^{\frac{1}{2}})^x$

$= (x^{1+\frac{1}{2}})^x$

$= x^{\frac{3}{2}x}$

95. If $9^{x-\frac{1}{2}} - 2^{2x-2} = 4^x - 3^{2x-3}$, then x is

यदि $9^{x-\frac{1}{2}} - 2^{2x-2} = 4^x - 3^{2x-3}$, तो x है

(a) $\frac{3}{2}$

(b) $\frac{2}{5}$

(c) $\frac{3}{4}$

(d) $\frac{4}{9}$